



Genetic and Evolutionary
Computation Conference



San José, Costa Rica
July 13-17, 2026

Evolving Multi-Channel Confidence-Aware Activation Functions for Missing Data with Channel Propagation

Naeem Shahabi Sani, Ferial Najiantabriz, Shayan Shafaei, Dean F. Hougen
School of Computer Science, University of Oklahoma
Norman, Oklahoma, USA
Shahabi@ou.edu

Motivation: Missing Data

Missing values are common in medical informatics, finance, and sensor networks. We handle them with imputation or masking, but both obscure a crucial distinction between what was measured and what was guessed.

Observed

Trusted, high-confidence measurement

Imputed

An estimate — uncertain, lower reliability

Standard scalar activations treat the two identically.

Fixed activations $f(x)$ (ReLU, Swish, ELU) operate only on the feature value x , with no missingness or confidence input.

Reliability can only be inferred implicitly, through the linear weights.

The Idea

Neuroevolution discovers neural components hard to design by hand. Prior work evolves scalar activations $f(x)$, but they remain blind to data quality. We extend the search to reliability-aware, multivariate activations.



Our approach: *evolve activations that respond to reliability and propagate that signal through the whole network.*

1

3C-EA

A GP framework that evolves multivariate activation functions depending on feature value, missingness, and confidence.

2

ChannelProp

A deterministic rule that propagates missingness and confidence through linear layers, so reliability survives beyond the input layer.

3

Empirical evidence

Experiments across natural and injected (MCAR / MAR / MNAR) missingness, with gains under structured, value-dependent missingness.

Method: Three Channels per Feature

$\tilde{x}$

Imputed value

Missing entries filled with training-set feature means preventing leakage, giving a uniform numeric input.

m

Missingness flag

Binary mask: 1 if missing, 0 if observed. Becomes a soft probability deeper in the network.

c

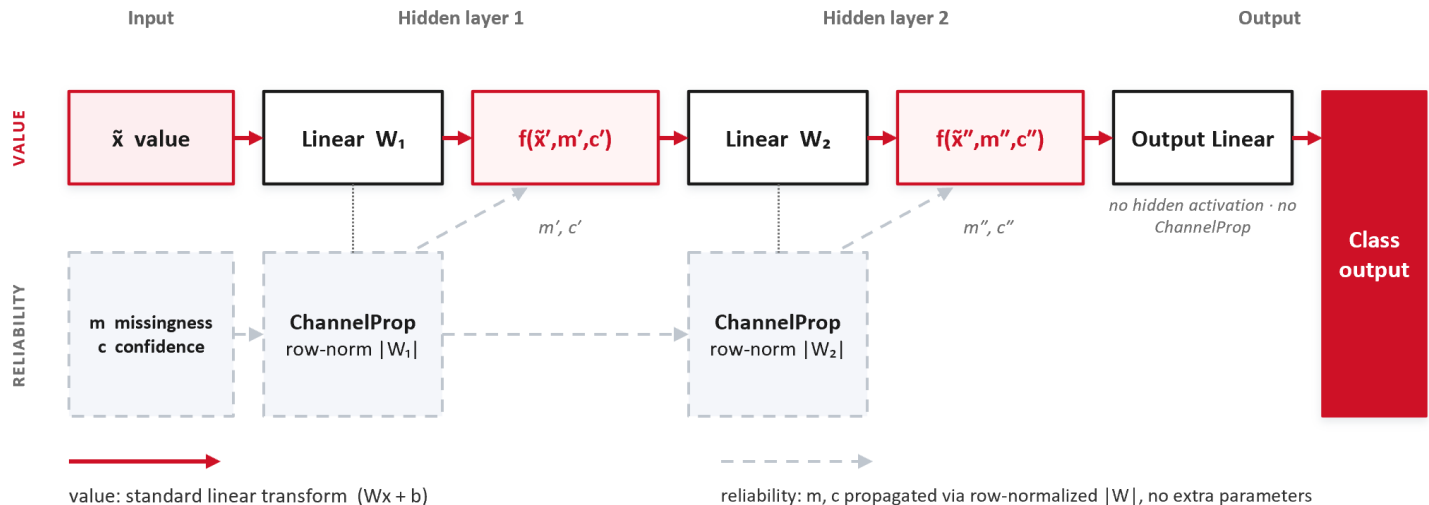
Confidence score

1.0 for observed. For imputed: $\max(\tau, 1 - r_j)$, where r_j = feature j 's missing rate in training set and $\tau = 0.1$ (lower bound). Rarely-missing features earn higher confidence.

A dataset-driven, feature-specific reliability heuristic with no extra learned parameters.

Method: The Three-Channel MLP (3C-MLP)

Each feature enters as three channels. The value channel is transformed by the linear layer; ChannelProp propagates missingness and confidence using the same layer's normalized weight magnitudes, so the evolved activation always receives current reliability information.



Shared Activation: every hidden layer applies the same evolved f . **Reliability Propagates:** m and c are propagated, not discarded after layer 1.

Method: ChannelProp

Mask-as-feature approaches lose reliability metadata after the first layer. ChannelProp carries missingness and confidence through every linear layer, using the layer's own weights to route reliability.

1. Routing matrix

$$A = |W| + \epsilon \quad (\text{larger weights route more influence})$$

2. Row-normalize

$$\tilde{A} = A / \sum A \quad (\text{each output row sums to 1})$$

3. Propagate

$$c_{\text{out}} = c_{\text{in}} \tilde{A}^T, \quad o_{\text{out}} = o_{\text{in}} \tilde{A}^T$$

4. Output missingness

$$m_{\text{out}} = 1 - o_{\text{out}}, \quad \text{clipped to } [0,1]$$

Why it matters

Deterministic

No new trainable parameters, depends only on the linear weights.

Soft signal

Binary at the input layer becomes continuous reliability deeper in the network.

Adaptive

Evolved activations can modulate their response when inputs are unreliable.

Method: Genetic Programming Search

Each activation is a symbolic expression tree over the three channels, evolved with a generational GP (fitness-proportional selection, subtree crossover, mutation, elitism).

Terminal & function sets

Terminals $\{ \tilde{x}, m, c \} + \text{constants} \{ 0, \pm 0.1, \pm 0.5, \pm 1, \pm 2 \}$

Unary 16 in total (tanh, σ , ReLU, ELU, softplus, exp, $\log|\cdot|$, sin, cos, ...)

Binary $+ - \times \div (\text{protected}) \min \max$

Fitness

$$F(T) = A_{val} + \lambda_d \cdot D - \lambda_s \cdot N - \lambda_h \cdot (H-1)$$

Validation accuracy, rewarding channel use (D) while penalizing node count (N) and depth (H) to curb bloat. Each candidate trains briefly inside a 3C-MLP; 3,000 candidates per run.

GP hyperparameters

Population	100
Generations	30
Max tree depth	3
Crossover prob.	0.7
Mutation prob.	0.15
Elite size	2

Experimental Setup

12 UCI datasets

7 naturally incomplete (Hepatitis, HouseVotes84, Heart Disease, Adult, ...) + 5 originally complete with injected missingness.

6 metrics, 30 runs

Accuracy, Precision, Recall, Specificity, F1, AUC reported as mean $\pm$ standard deviation over 30 seeds.

3 mechanisms

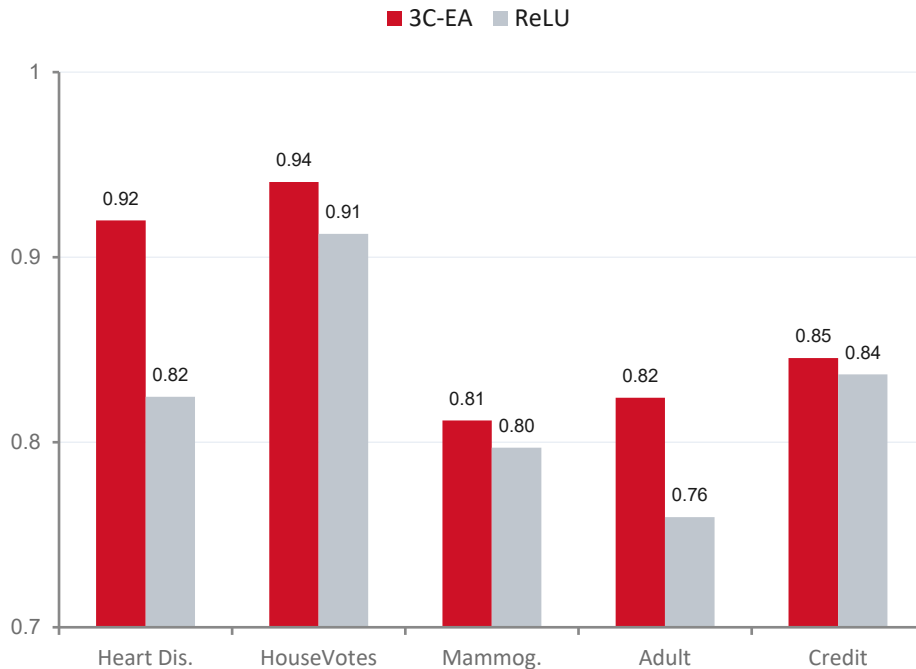
Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR), injected at controlled rates.

Fair baselines

ReLU, Swish, LeakyReLU, ELU — identical splits, preprocessing & architecture; only the activation differs.

Significance: one-sided Wilcoxon signed-rank test, H_1 : 3C-EA > baseline ($p < 0.05$ / $p < 0.01$).

Results: Natural Missingness



Best accuracy on 6 of 7 datasets (5 shown).

Highlights

+9 pts

accuracy on Heart Disease vs. ReLU
(0.920 vs. 0.825).

0.55 vs 0.26

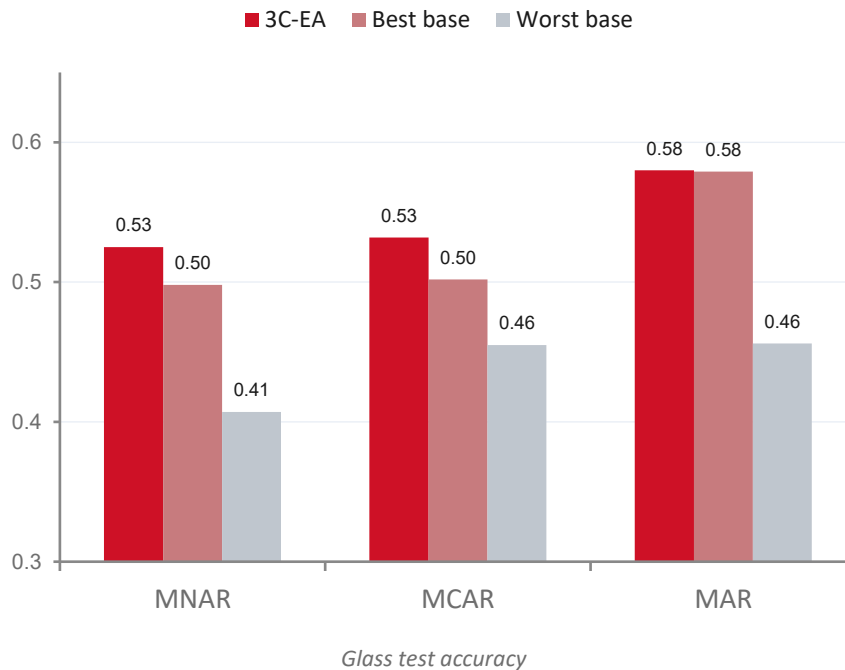
Cylinder Bands specificity (38%
missing).

Recall 0.54

Adult (48,842 rows) vs. 0.01, no
majority collapse.

Results: Injected Missingness

With 20% injected missingness on originally complete data, the advantage is largest on the multi-class, imbalanced Glass dataset across all three mechanisms.



Where reliability helps

MNAR

Hardest case. AUC 0.77 vs 0.65–0.73.

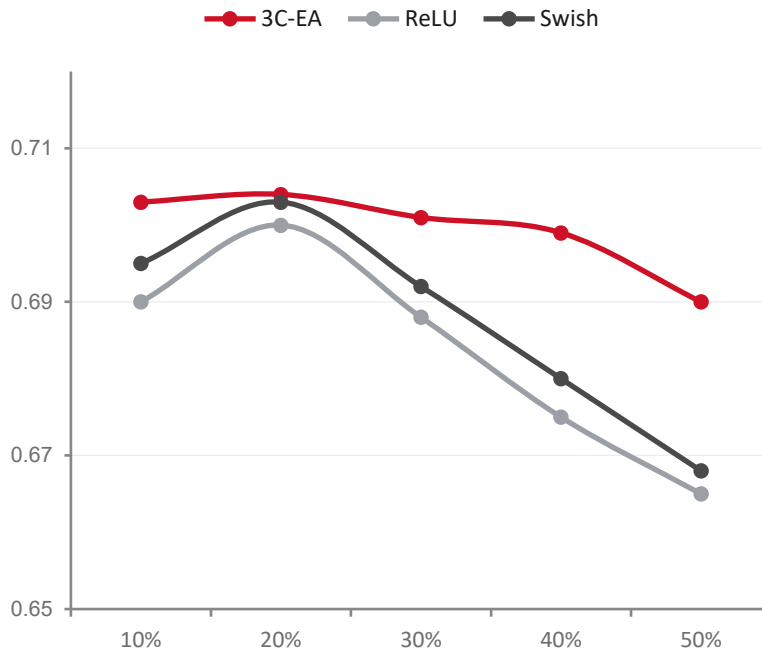
PIMA recall

Rises sharply avoids majority collapse.

Mushroom

98.5% acc, lowest variance stable at scale.

Results: Robustness & Ablation



PIMA accuracy vs. missing rate; gap widens as quality drops

Ablation — Glass, MNAR @ 40%

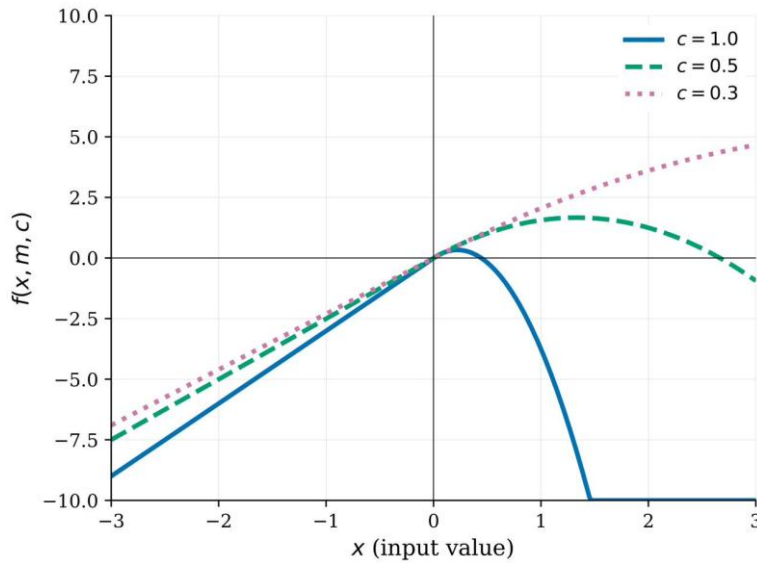
Variant	Acc.
Full $f(\tilde{x}, m, c)$	0.435
No confidence $f(\tilde{x}, m)$	0.407
No flag $f(\tilde{x}, c)$	0.386
No ChannelProp	0.381

Removing any component degrades performance.

What the Evolved Functions Learned

Confidence reshapes the response

On Heart Disease, the activation is richly nonlinear at high confidence ($c = 1.0$) but flattens toward conservative, near-linear behavior at low confidence ($c = 0.3$).



Amplifying uncertain inputs

On Glass: $f = \min(x \cdot m, x) + x$.

Imputed ($m=1$) $\rightarrow 2x$ always. Observed ($m=0$) $\rightarrow x$ for $x \geq 0$, $2x$ for $x < 0$ — doubling imputed values to counter their uncertainty.

Selective channel use

Across 30 runs on Hepatitis:

- Most winners use missingness or confidence channels.
- Some competitive solutions rely on x alone.
- The diversity reward encourages not forces channel use: a channel is kept only when it earns its validation performance.

Conclusions

3C-EA

Evolving activations on $(\tilde{x}, m, c)$ is competitive with, and often beats, fixed baselines.

ChannelProp

Deterministically carries reliability through the network, with no extra parameters.

Biggest wins

Structured / value-dependent missingness, high missing rates, multi-class tasks.

Future work

Extend ChannelProp to CNNs, Transformers, LLMs · treat MCAR / MAR / MNAR distinctly · combine heuristic with learned confidence.

Thank you

Questions welcome

Naeem Shahabi Sani · shahabi@ou.edu · University of Oklahoma



Code on GitHub